

CG Height Test Plan

Method A: Side Tilt (Balance) Test

1 Purpose

This document outlines how we find the center of gravity (cg) height of ARG26 through a physical test. Although we can make reasonable estimates through CAD, validating the cg height is important because it influences a lot of our load transfer and handling characteristics.

2 Method A: Side Tilt (Balance) Test

2.1 Theory

A body balances on a pivot when its CG is directly above the pivot. In this test the car is tipped sideways onto the tires of one side, so it rests only on the outer edges of those two tires. The car is slowly rotated until it balances and can be held in place with a couple of fingers, with the crane strap slack. At that point the CG lies on the vertical line through the pivot.

In the car's own frame, the CG sits half the car's width, $d/2$, in from the pivot edge and a height h above the ground plane. When the car is rotated by the balance angle θ , the line from the pivot to the CG is vertical, so $\tan \theta = (d/2) / h$.

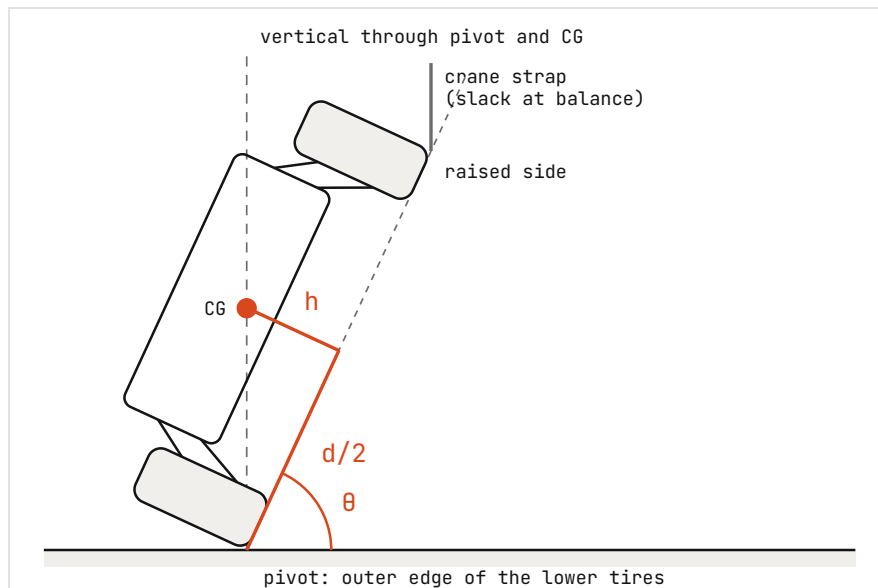


FIGURE 1 Front view of the car balanced on its left tires. At balance the CG is directly above the pivot, so $\tan \theta = (d/2) / h$.

2.2 Materials

#	Item	Where
1	ARG26	Lab
2	Driver	—
3	Vehicle scales (corner weights)	Lab
4	Digital level	Lab
5	Crane with strap	Lab
6	Dummy shocks and dummy cones	Lab
7	Magnesium rims	Lab
8	Tire pump and pressure gauge	Lab
9	Tape measure and straight edge	Lab
10	Plumb bob	Lab
11	Wheel chocks for the lower tires	Lab
12	Camera or phone	—

2.3 Pre-Test Procedure

1. Install the dummy shocks and dummy cones.
2. Install the accumulator and bolt it down (ask someone on high voltage).
3. Make sure the car is on the magnesium rims.
4. Overinflate the tires to 30 psi.

2.4 Measurements Before Tilting

Record these with the car level on the scales, driver in the car.

Symbol	Measurement	Value
W_{FL}	Front left corner weight (lb)	
W_{FR}	Front right corner weight (lb)	
W_{RL}	Rear left corner weight (lb)	
W_{RR}	Rear right corner weight (lb)	
W	Total weight (lb)	
L	Wheelbase (in)	
d_f	Front width, outside tire edge to outside tire edge (in)	
d_r	Rear width, outside tire edge to outside tire edge (in)	
p	Tire pressure (psi)	

Symbol	Measurement	Value
θ_0	Level reading, car level (deg)	

2.5 Equations

The front and rear of the car have different widths, so the width is taken at the lengthwise position of the CG (Figure 2) [3].

The CG sits a distance a behind the front axle, set by the fraction of weight on the rear axle, $W_R = W_{RL} + W_{RR}$:

$$a = L \times W_R / W$$

$$d = d_f + (d_r - d_f) \times W_R / W$$

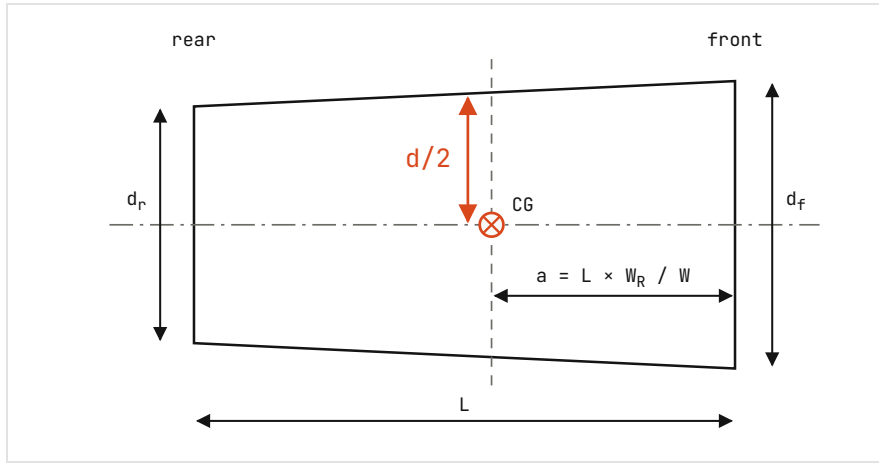


FIGURE 2 Plan view. The width d is taken across the outside tire edges at the lengthwise position of the CG. Adapted from [3].

CG height above the ground at static ride height:

$$h = (d / 2) / \tan \theta$$

where $\theta = |\theta_{\text{bal}} - \theta_0|$ is the change in the level reading from level to balanced. This assumes the CG is on the car's centerline; averaging tilts onto both sides cancels out a small side-to-side offset.

An error in θ changes h by roughly $\Delta h \approx (d / 2) \times \Delta \theta / \sin^2 \theta$ ($\Delta \theta$ in radians). A 0.2° error gave roughly a 0.1 in error in h for the car measured in [3], so measure the balance angle carefully and repeat it.

2.6 Procedure

1. Complete the pre-test procedure and the measurements before tilting.
2. Chock the lower-side tires so they cannot slide.
3. Attach the crane to the raised side and lift slowly. Spotters hold the car at all times.
4. Keep raising until the car balances on the outer edges of the lower tires, holding in place with a couple of fingers and the strap slack.
5. Confirm balance with the plumb bob, then record the level reading at the same spot, θ_{bal} .
6. Lower the car back to level. Repeat for three trials.

7. If safe, repeat three trials tilting onto the other side.

2.7 Data Sheet

Trial	Pivot Side	θ_{bal} (deg)	θ (deg)	h (in)
A1				
A2				
A3				
A4				
A5				
A6				

2.8 ARG26 Results

Result	Value
CG height, average of all trials (in)	
Lowest trial (in)	
Highest trial (in)	
Spread, highest minus lowest (in)	

To be filled out after testing.

3 References

1. J. Segers, Analysis Techniques for Racecar Data Acquisition, 2nd ed., pp. 222–224.
2. W. F. Milliken and D. L. Milliken, Race Car Vehicle Dynamics, SAE, 1995, Section 18.2.
3. J. Hakewill, "Measuring Center-of-Gravity Height on a Formula Car," June 2007. Method credited to Claude Rouelle.
4. J. Callister and L. Sganderlla, ARG16 CG Height Measurement Testing Protocol, Cornell Racing.
5. CG Height Test, Suspension TRR, Cornell Racing, September 2026.